



PHOTO FRAME GS 01

E-Ink Display 960x540px [LilyGo T5 E-paper / ESP32 WROVER E]

E-ink digital picture frame with remote image updates via FTP and configuration settings adjustable through Telnet. Features deep sleep mode for extended battery life.

OPERATING MODES

1 PHOTO FRAME MODE (Normal Operation)

- Automatically displays JPG images from **/images/** folder
- Cycles through images based on configured wake interval
- **Supports:** Seconds, Minutes, Hourly, Half-Day, Daily, Weekly, Monthly intervals
- Deep sleep between image changes for battery conservation
- Low battery detection with visual warning

2 MAINTENANCE MODE (Configuration & File Management)

Activation: Hold Maintenance Button (GPIO 39) for 3 seconds during boot

Features available:

- WiFi connection (AP or STA mode)
- Telnet server for remote configuration
- FTP server for file upload/download
- mDNS support for easy hostname access

Exit: Hold Maintenance Button for 3 seconds to reboot

Factory Reset: Hold Reset Button (GPIO 34) for 30 seconds

3 LOW BATTERY MODE (Empty battery)

- Triggered when battery voltage is too low
- Displays battery icon with voltage and percentage
- Enters extended deep sleep to preserve battery

CONNECTION & ACCESS

WIFI AP MODE (Default)

SSID: PhotoFrameGS01
Password: 123456789
IP: 192.168.4.1
Gateway: 192.168.4.1
Subnet: 255.255.255.0

WIFI STA MODE

sta_ssid – WiFi name
sta_password – WiFi password
sta_enable – static IP
sta_ip/gateway/subnet/dns
Set: ap_enable = false

mDNS (optional)

mdns_enable = true
mdns_hostname = photoframegs01
Access: photoframegs01.local

TELNET ACCESS

Port: 23
User: admin
Pass: 123456789
Session: 1 hour

FTP ACCESS

Port: 21
User: admin
Pass: 123456789

TELNET COMMANDS

SYSTEM

help – command list
clear – clear screen
reboot – restart
reset config – factory reset
exit – disconnect
logout – re-auth

INFORMATION

meminfo – RAM usage
netinfo – IP/MAC/DNS
fsinfo – storage usage
sketchinfo – firmware size
nvsinfo – NVS usage
batinfo – battery info
date – current time
timestamp – Unix epoch

FILES

list – show all files
cat <filename> – view content

CONFIGURATION

config <key> – get configuration
config <key> [value] – set configuration

DOWNLOAD

fetch <url> [name] – HTTP/HTTPS
– max 200KB, .j(e)pg

FTP & CONFIGURATION

FILE STRUCTURE

/config.ini – settings
/images/ – photos

IMAGE REQUIREMENTS

Format: JPG/JPEG
Size: 960x540px
Max.: 200 KB per file
Tip: Grayscale for best

magic photo.jpg --colorspace Gray --contrast-stretch 255 --noindex-dither about 1k pixel.jpg

CONFIG.INI SECTIONS

[device] – name, version
[display] – jpg settings
[ap mode] – AP settings
[sta mode] – WiFi creds
[static ip] – IP settings
[ntp] – time server
[telnet] – telnet config
[ftp] – ftp config
[timer] – wake interval

WAKE UP VALUES

1=Sec, 2=Min, 3=Hour,
4=12h, 5=Day, 6=Week, 7=Month

BUTTONS

Next Image (1s), Reset: (Factory Reset 30s),
Wake/Maintenance (1s)

DESIGNED BY  FROM 

BUTTONS

NEXT
IMAGE

RESET

MAIN-
TENANCE

USB-C CONNECTOR